



SECURE YOUR FINANCIAL FUTURE

ifss.

ITR FILING SUGGESTION SOFTWARE

PROJECT CODE : IFSS

TEAM : SUJAL KAWARE

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Project Insight

PS : USERS REQUIRE A PLATFORM WHERE THEY CAN STORE ALL THEIR FINANCIAL VALUE DATA AND A SIMPLE CLICK WOULD PROVIDE THEM WITH THEIR INCOME ANALYSIS AND TAX FILING SUGGESTIONS

ARCHITECTURE

- FRONTEND: NEXT.JS/REACT WITH SHADCN UI
- BACKEND: CPP FOR COMPUTATION AND, FLASK FOR APIS AND BACKEND SERVER HOSTING
- DATABASE: MONGODB FOR PERSISTENT STORAGE

TECH STACK

FRONTEND:

NEXT.JS 15.2.2, REACT 19.0.0, RADIX-CONS, JSON-WEBTOKEN, ZOD-RESOLVER, CUSTOMIZED SHADCN UI, TAILWIND CSS,

BACKEND:

CPP, FLASK2.0.2, FLASK CORS 3.0.10, NLOHMANN, CATCH2

TESTS

BACKEND:

CATCH2 FOR CPP TESTS

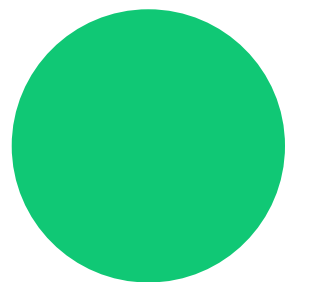
FRONTEND:

JEST FOR UNIT TESTS CONFIGURATION, AND ES LINT FOR LINTERS

DEPLOYMENT & MONITORING

CI : GITHUB ACTIONS AND MAKE

CD: DOCKER CONTAINER



features

VERIFIED PAN LOGIN SYSTEM

**INCOME DETAILS SAVING AND
INCOMETAX CALCULATIONS &
ANALYSIS**

**AI CHATBOT FOR USER HELP AND
AI GENERATED TAX SUMMARY**

INBOX AND MESSAGING SYSTEM

CA MANAGEMENT SYSTEM

MONGODB CONNECTED USER DATABASE FOR REMOTE DATA ACCESS

**AUTHENTICATION AND VERIFIED LOGIN SYSTEM FOR USER, CA AND ADMIN
INCLUDING EMAIL OTP VERIFICATION**

USER FRIENDLY DASHBOARD AND PROFILE EDITING FEATURES

INCOME TAX CALCULATIONS AND INCOME TAX ANALYSIS.

**FLASK (PYTHON) SERVER FOR INTEGRATING CPP BACKEND AND NEXTJS
FRONTEND AND PASSING API CALLS. WITH CORS (CROSS-ORIGIN RESOURCE
SHARING) FOR CONTROLLED ACCESS TO RESOURCES HOSTED ON DIFFERENT
DOMAINS (PASSING JSON FILES)**

AI CHATBOT INTEGRATION FOR USER ASSISTANCE

AI SUMMARIZED DEDUCTIONS AND TAX SUMMARY, SUGGESSTIONS

CA REGISTRATION AND CA APPROVAL BY ADMIN, WITH EMAIL NOTIFICATION.

**WORK ALLOTMENT TO CAS BY ADMIN BASED ON JOB SEQUENCING AND
AVAILABILITY**

**INBOX SYSTEM FOR COMMUNICATIONS BETWEEN USERS AND ALLOTTED CAS AND
ADMIN TO CAS.**

Challenges

SECURITY OF USER DATA WAS A MAJOR CONCERN , HENCE WE OPTED FOR A SUGGESTION SOFTWARE

CROW SERVER : AT EARLY STAGES OF DEVELOPMENT WE USED CROW SERVER FOR LINKING THE NEXTJS FRONTEND WITH CPP BACKEND, WE HAD SOME SUCCESS IN IT FOR PASSING ALERTS AND MESSAGES .

HOWEVER IT FAILED CORS : CROSS ORIGIN RESOURCE SHARING AND DATA WASNT PASSED AS NEEDED
TRIED DEBUGGING ON IT FOR 3 DAYS , SHIFTED TO FLASK PYTHON SERVER

CONTINUOUS INTEGRATION : DONE BY GITHUB ACTIONS CI.YML

CONTINUOUS DEPLOYMENT : USE OF DOCKER TO COMBINE TWO SERVERS OF PYTHON
BACKEND AND NPM FRONTEND DOWNLOADING DEPENDENCIES AND REQUIREMENTS

Scalability

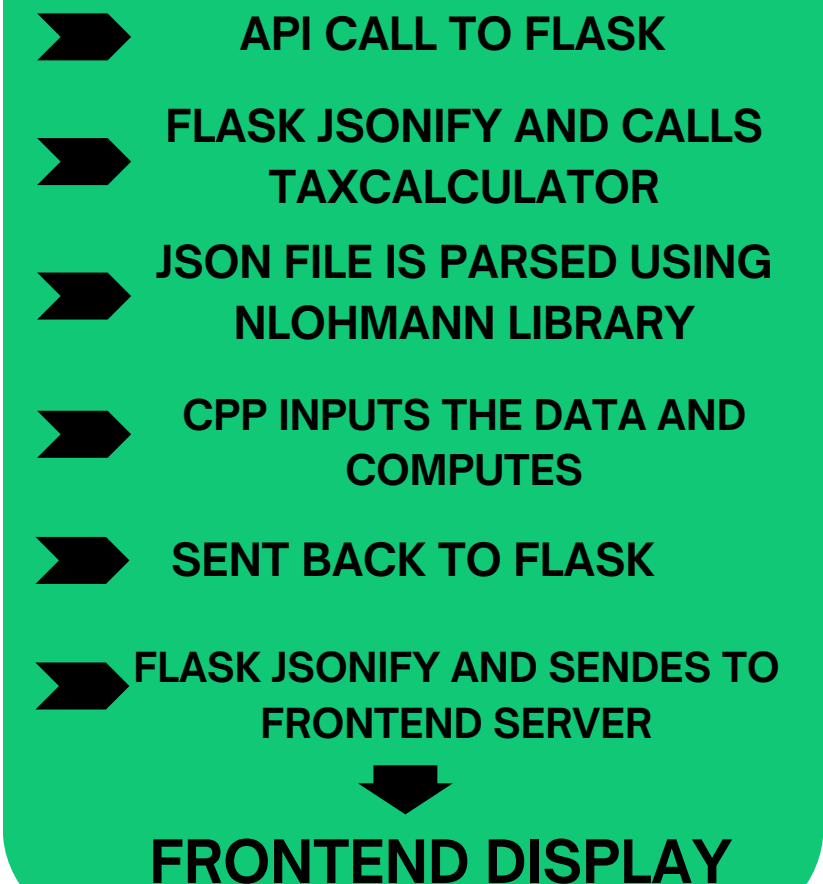
OPTIMISING DATABASE SECURITY
IMPLEMENTING SCALABLE DATABASE

USE OF SCALABLE DATABASE
USE OF GUNICORN AS PYTHON SERVER

EXTENDING SUPPORT TO ITR 2,3
DIRECT TRANSFER OF DATA TO ITR FORMS

CURRENT APPROACH

FRONTEND FORMDATA





Thank You

Please feel free to ask any questions or provide feedback.

Contact us: ifsspro@gmail.com